

Home Tasks - 01

Advanced Database Management Systems — Home Assignment

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Q1. Candidates With All Required Skills

Given a table of candidates and their skills, you're tasked with finding the candidates best suited for an open Data Science job. You want to find candidates who are proficient in Python, Tableau, and PostgreSQL.

Write a query to list the candidates who possess all of the required skills for the job. Sort the output by candidate ID in ascending order.

Assumption:

- There are no duplicates in the candidates table.

candidates Table

Column Name	Type
candidate_id	integer
skill	varchar

candidates Example Input

candidate_id	skill
123	Python
123	Tableau
123	PostgreSQL
234	R
234	PowerBI
234	SQL Server
345	Python
345	Tableau

Example Output

candidate_id
123

Solution

```
SELECT CANDIDATE_ID  
FROM CANDIDATES  
WHERE SKILL IN ('Python', 'Tableau', 'PostgreSQL')  
GROUP BY CANDIDATE_ID  
HAVING COUNT(SKILL) = 3  
ORDER BY CANDIDATE_ID ASC;
```

Q2. Facebook Pages With Zero Likes

Assume you're given two tables containing data about Facebook Pages and their respective likes (as in "Like a Facebook Page").

Write a query to return the IDs of the Facebook pages that have zero likes. The output should be sorted in ascending order based on the page IDs.

pages Table

Column Name	Type
page_id	integer
page_name	varchar

pages Example Input

page_id	page_name
20001	SQL Solutions
20045	Brain Exercises
20701	Tips for Data Analysts

page_likes Table

Column Name	Type
user_id	integer
page_id	integer
liked_date	datetime

page_likes Example Input

user_id	page_id	liked_date
111	20001	04/08/2022 00:00:00
121	20045	03/12/2022 00:00:00
156	20001	07/25/2022 00:00:00

Example Output

page_id
20701

Solution

```
SELECT P.PAGE_ID
FROM PAGES AS P
LEFT JOIN PAGE_LIKES AS L ON P.PAGE_ID = L.PAGE_ID
WHERE L.USER_ID IS NULL
ORDER BY P.PAGE_ID ASC;
```

Q3. Unfinished Parts in Assembly

Tesla is investigating production bottlenecks and they need your help to extract the relevant data. Write a query to determine which parts have begun the assembly process but are not yet finished.

Assumptions:

- parts_assembly table contains all parts currently in production, each at varying stages of the assembly process.
- An unfinished part is one that lacks a finish_date.

This question is straightforward, so let's approach it with simplicity in both thinking and solution.

Effective April 11th 2023, the problem statement and assumptions were updated to enhance clarity.

parts_assembly Table

Column Name	Type
part	string
finish_date	datetime
assembly_step	integer

parts_assembly Example Input

part	finish_date	assembly_step
battery	01/22/2022 00:00:00	1
battery	02/22/2022 00:00:00	2
battery	03/22/2022 00:00:00	3
bumper	01/22/2022 00:00:00	1
bumper	02/22/2022 00:00:00	2
bumper		3
bumper		4

Example Output

part	assembly_step
bumper	3
bumper	4

Solution

```
SELECT PART, ASSEMBLY_STEP
FROM PARTS_ASSEMBLY
WHERE FINISH_DATE IS NULL;
```